Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
BCA | Sem-3 | Winter-2022

Course Code	:	2104CS301	**Date**	:	17-10-2022
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	List advantages & disadvantages of computer network.	**4**
	(B)	Show the usage of Computer Network.	**3**
		OR	
		Define computer network with its characteristics.	
	(C)	Explain types of computer network (LAN, MAN, WAN) with example.	**7**
		OR	
		Classify Computer Network Architecture in brief.	
Q.2	**(A)**	Describe Physical Layer.	**4**
	(B)	Draw OSI Reference Model.	**3**
		OR	
		Draw TCP/IP Model.	
	(C)	Compare OSI Reference Model and TCP/IP Model.	**7**
		OR	
		Explain functions of Network Layer.	
Q.3	**(A)**	Discuss fiber optic cable with suitable diagram.	**4**
	(B)	Illustrate the Bluetooth Applications used in real life.	**3**
		OR	
		Explain Infrared.	
	(C)	Explain Bluetooth Protocol Stack with diagram.	**7**
		OR	
		Describe radio wave.	

Q.4	**(A)**	Compare switch and router.	**4**
	(B)	Define networking device. List the devices used in networking.	**3**
		OR	
		Write a short note on LAN card.	
	(C)	Carry out functionality of router in networking.	**7**
		OR	
		Carry out functionality of switch in networking.	
Q.5	**(A)**	List IP address Classes and range of each class.	**4**
	(B)	Write a short note on SMTP.	**3**
		OR	
		Show the behavior of non-persistent types HTTP with request-response message.	
	(C)	Given IP address 188.25.45.48/20. Find below details: a) Class b) Subnet mask c) What are the total subnets? d) What are the total hosts?	**7**
		OR	
		Given IP address 192.168.200.5/25. Determine below details: a) Class b) Subnet mask c) What are the total subnets? d) Give the first host ID and the last host ID of each subnet.	

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